Federated Learning and Heterogeneous Data

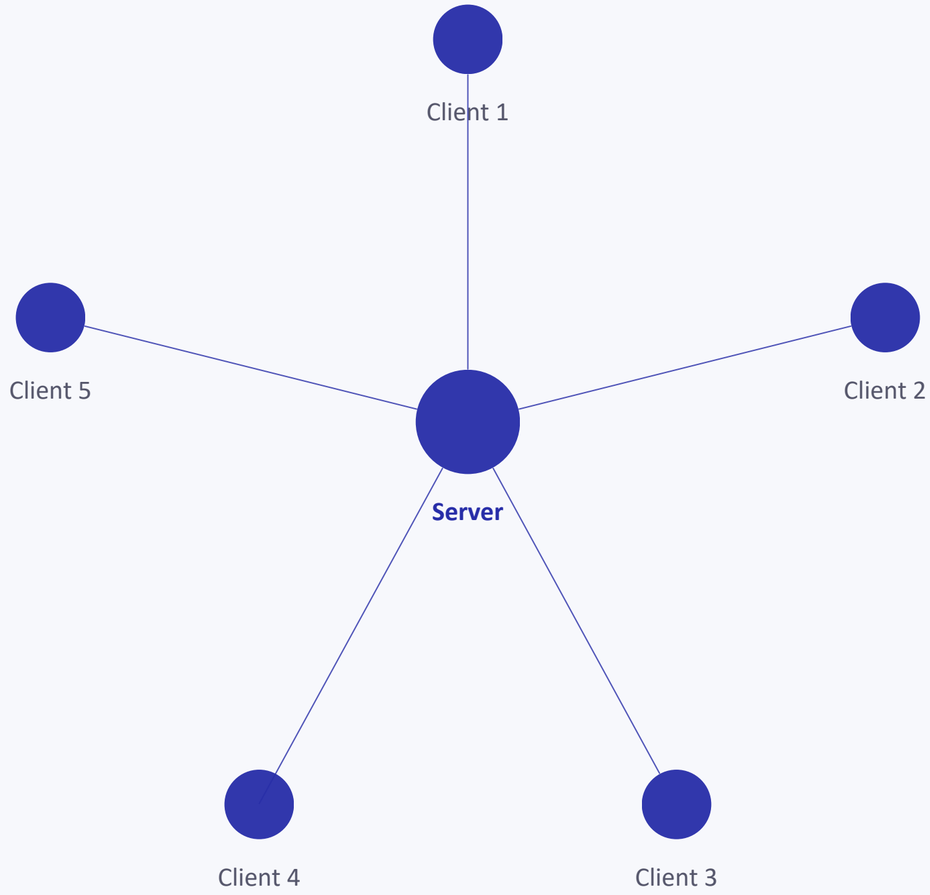
FedAvg and FedProx compared on IID and Non-IID distributions

Signal Processing for Machine Learning

Alessandro Querqui • 18/09/2026

Why data heterogeneity is a problem

- **Federated Learning**: many devices train a model locally and share only the updates with a central server
- The server aggregates the local models (FedAvg) to build a global model
- Implicit assumption: client data are fairly similar to each other (IID)
- In reality, each user/device generates different data (Non-IID): **statistical heterogeneity**



Research question

How much does FedAvg degrade when this assumption breaks down, and are known mitigations (FedProx) enough to fix it?

The optimization view: FedAvg vs FedProx

FedAvg — local step

$$min_w \; F_k(w)$$

Each client purely minimizes its own local loss. Nothing keeps it from drifting toward its own local optimum.

FedProx — local step

$$min_w \; h_k(w) = F_k(w) + \boldsymbol{(\mu/2)} \; \|w - w^t\|^2$$

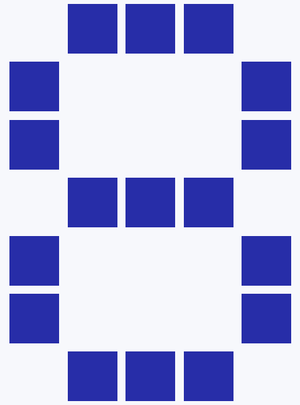
An extra proximal penalty is added: it grows with the squared distance between the local weights w and the global weights w^t received at the start of round t.

μ is the knob: $\mu = 0$ recovers plain FedAvg; a larger μ pulls local updates back toward w^t, trading local adaptation for global consistency (exactly the mechanism tested in the Extreme and Dirichlet scenarios ahead).

Experimental setup

Dataset	MNIST Digit Dataset | 10 classes, 28×28 images
Model	Softmax regression (multinomial logistic regression)
Clients	5
Global rounds	100
Local epochs	20 (full-batch gradient descent)
Local learning rate	0.5
Total samples	10000 | Stratified 80/20 Train/Test split

Feature standardization (per-pixel z-score) before training



MNIST digit 8

Model

$$\hat{y} = softmax(\boldsymbol{W}x + \boldsymbol{b})$$

Linear scores per class, turned into probabilities via softmax.

Three scenarios, three degrees of heterogeneity

IID

Each client receives a random, representative sample of all 10 digits.



example: class distribution for a single client

Tested with: FedAvg

Extreme Non-IID

Each client sees ONLY 2 out of 10 digits. Zero information about the other 8.

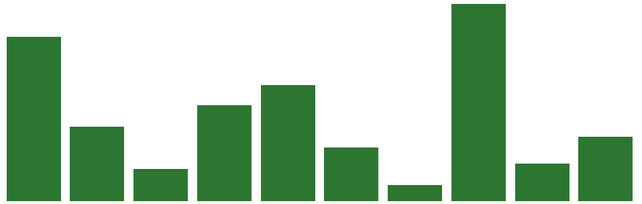


example: class distribution for a single client

Tested with: FedAvg + FedProx

Moderate Non-IID (Dirichlet, α=0.5)

Each client sees ALL digits, but in highly unbalanced proportions.



example: class distribution for a single client

Tested with: FedAvg + FedProx

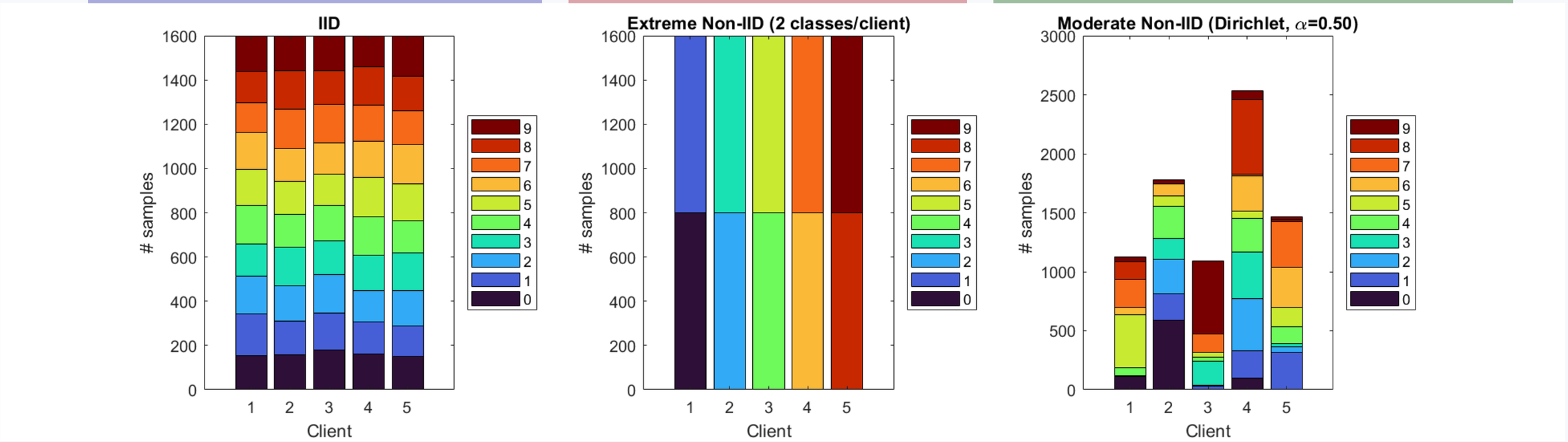
How clients see the data

Each panel shows, for the 5 clients, how many images of each digit (0-9) they hold

IID: bars are nearly identical across clients

Extreme: each client has only 2 classes in its bar

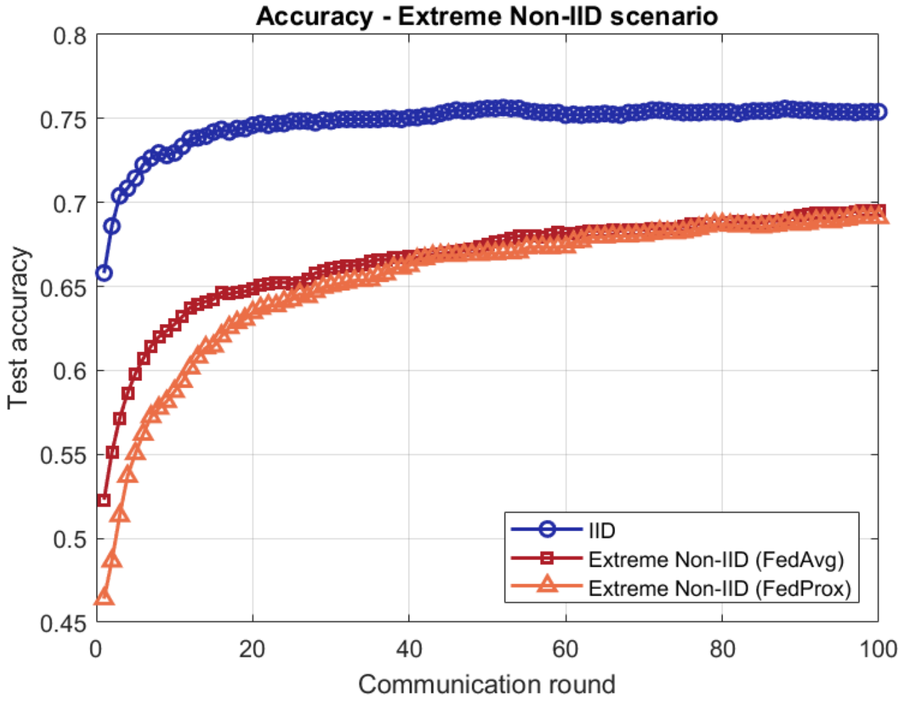
Dirichlet: all 10 classes present everywhere, but in very different proportions

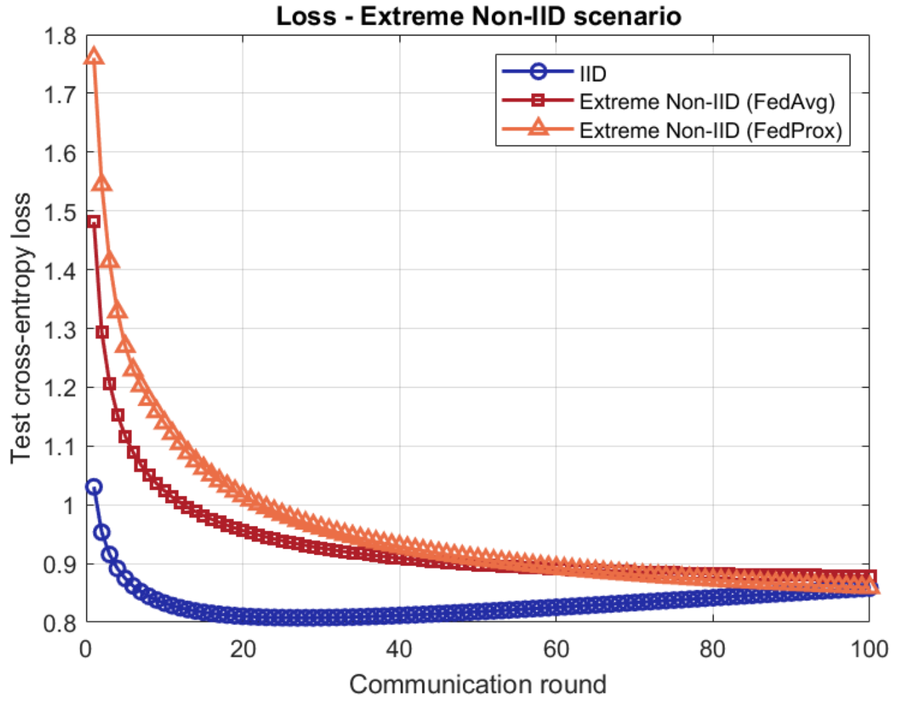


Client label distributions (IID / Extreme Non-IID / Dirichlet)

FedAvg: IID vs Extreme Non-IID

- Final accuracy: IID 0.754 vs Extreme Non-IID 0.695: gap persists throughout training
- The Non-IID model starts more unstable (oscillations in the early rounds) and never fully closes the gap

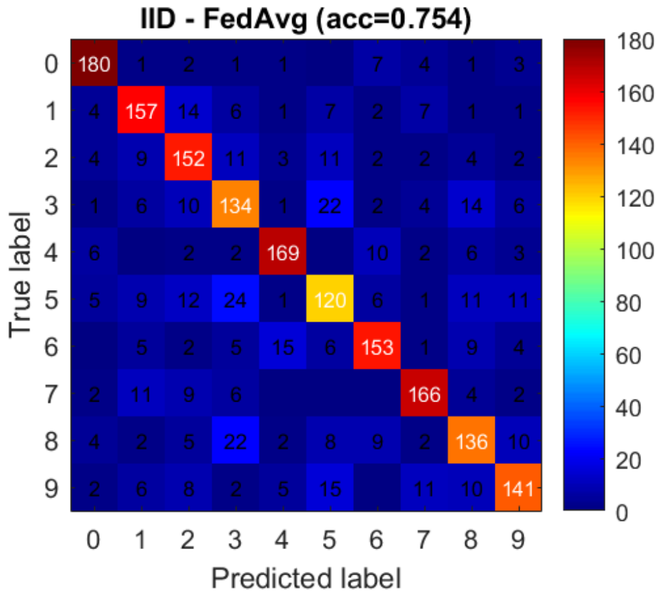


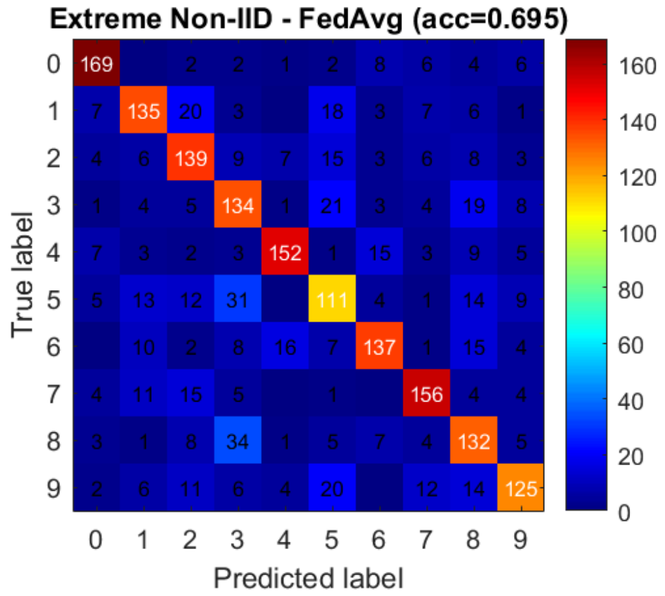


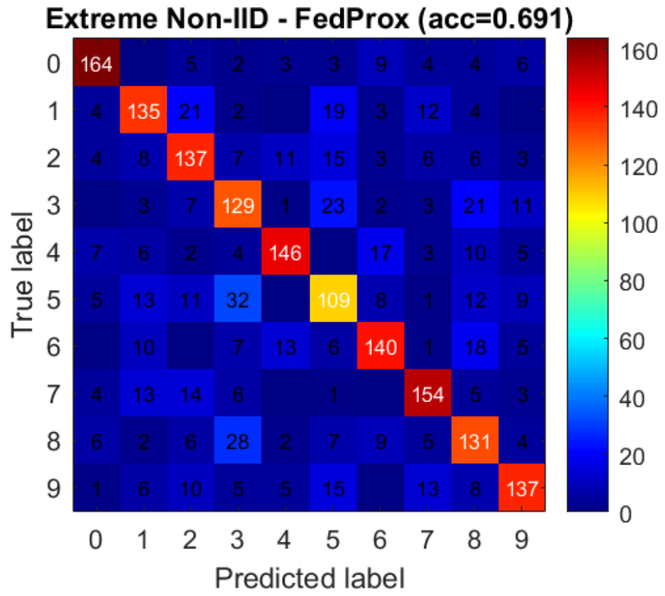
Accuracy and Loss — Extreme Non-IID scenario (IID vs FedAvg vs FedProx)

Why FedProx doesn't help in the extreme case

- FedProx: 0.691 accuracy; practically identical to FedAvg (0.695), no real benefit
- FedAvg and FedProx confusion matrices are nearly indistinguishable: same errors, same intensity
- Cause: the proximal term limits how far a client drifts from the global model, but it cannot invent information the client never saw (8 out of 10 digits missing)



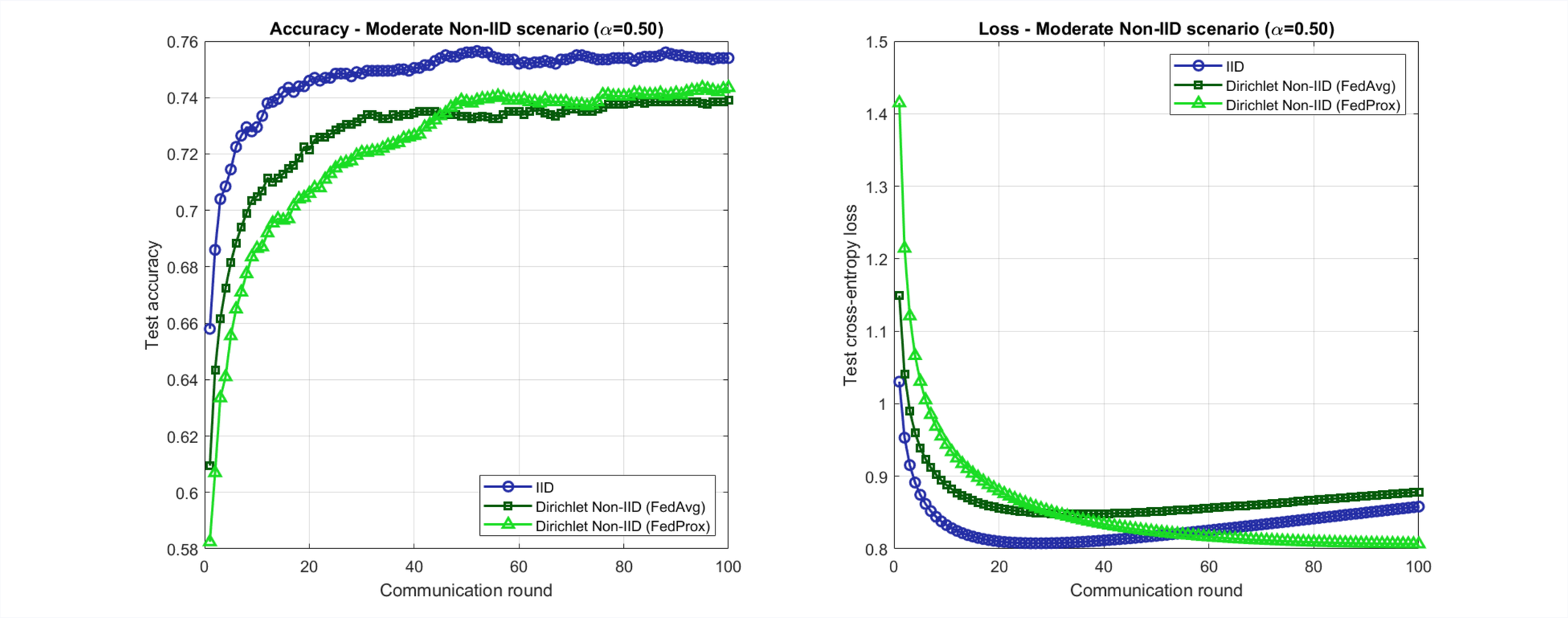




Confusion matrices — IID / Extreme FedAvg / Extreme FedProx

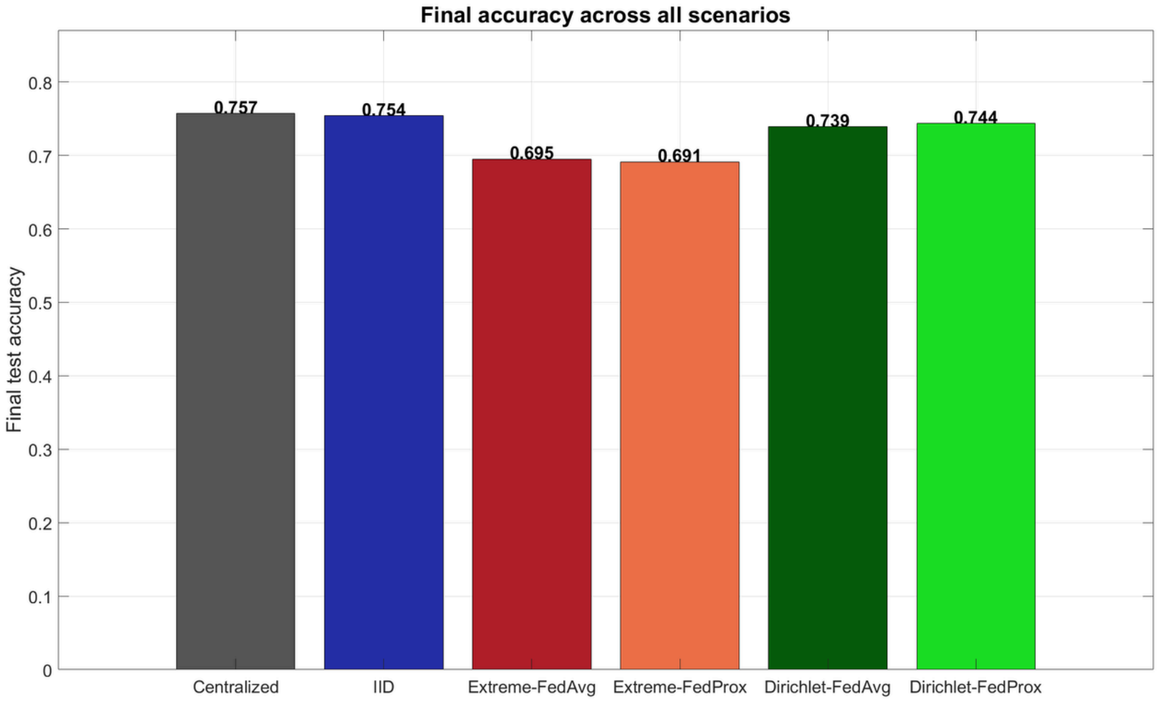
Moderate Non-IID (Dirichlet): here FedProx helps

- Final accuracy: FedAvg 0.739 → FedProx 0.744 (+0.005, small but consistent)
- Stronger signal in the LOSS: FedAvg rises again after round 20-30 (drift accumulates over time), FedProx stays monotonically decreasing throughout training
- Here the problem really is client drift: exactly what FedProx is designed to correct



Accuracy and Loss — Dirichlet Non-IID scenario (IID vs FedAvg vs FedProx)

Summary: final accuracies



Summary bar chart — final accuracy across the 5 scenarios (plus centralized one)

Conclusions

- *FedAvg converges well on IID data, but degrades on Non-IID data*
- *The severity of the degradation depends on the TYPE of heterogeneity, not just its presence*
- *FedProx is a targeted mitigation: effective against client drift (Dirichlet), not against structurally missing information (extreme non-IID)*

Limitations



- Single seed/run: no averaging across runs
- Only one α value tested for the Dirichlet case
- muProx set heuristically
- Linear model (softmax regression): findings may not transfer to deep networks

Future work



- Validation across multiple seeds to estimate result variance
- Continuous α sweep (Dirichlet) for heterogeneity → accuracy curve
- muProx tuned via systematic search
- Extension to a non-linear model (multi-layer neural network)